

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE CODE: MLA03 COURSE NAME: Reinforcement learning for Environment and Agent

COURSE OUTCOME COVERED

***CO5:** Implement policy-based reinforcement learning methods to develop efficient solutions for complex environments. (BL2, BL3, BL6)*

Assessment Tool Weightage: 30%

Dr G. A. SENTHIL

QUESTIONS: 10

Total Marks: 50

Annexure A

Analytical Problem-Solving Assessment

Duration: 1hr

1. A warehouse robot must determine the shortest path to transport goods while avoiding obstacles. Apply **Dynamic Programming** to develop an optimal policy and explain how value iteration can improve navigation efficiency. **(5 Marks)**
2. A recommendation system collects customer interactions over multiple sessions. Apply **Monte Carlo Control** to update the policy and explain how the agent improves recommendations with experience. . **(5 Marks)**
3. A delivery robot operates in an uncertain environment. Apply the **SARSA** algorithm to update the action-value function and explain how the learned policy adapts to environmental changes. . **(5 Marks)**
4. A robotic arm performs continuous object manipulation. Apply the **REINFORCE** algorithm to develop a policy that maximizes task completion accuracy. .

(5 Marks)

5. An autonomous vehicle is trained using **Q-Learning** and **Deep Q-Networks (DQN)**. Analyze the differences in learning performance, convergence speed, and scalability for complex driving environments. . (5 Marks)
6. A game-playing AI is implemented using **DQN**, **Double DQN (DDQN)**, and **Dueling DQN**. Analyze the advantages of DDQN and Dueling DQN over the standard DQN architecture. . (5 Marks)
7. A smart drone navigation system uses **A2C** and **A3C** algorithms. Analyze the impact of parallel learning on convergence speed, computational efficiency, and policy stability. . (5 Marks)
8. A healthcare decision-support system operates with incomplete patient information. Analyze how a Partially Observable Markov Decision Process (POMDP) improves decision-making compared to a standard MDP. . (5 Marks)
9. A robotic navigation system is trained using PPO, TRPO, and DDPG. Evaluate the suitability of each algorithm for continuous control tasks based on stability, sample efficiency, and computational complexity. . (5 Marks)
10. A smart city traffic management system uses **Multi-Agent Reinforcement Learning (MARL)** to coordinate traffic signals. Evaluate the **effectiveness of MARL** in improving traffic flow while considering scalability, coordination, fairness, and ethical challenges. . (5 Marks)

1. Warehouse Robot – Dynamic Programming

Dynamic Programming can find the shortest path by dividing navigation into smaller states. Each grid cell represents a state and possible movements are actions. Value Iteration repeatedly updates the value of each state using the Bellman equation:

$$[V(s) = \max_a [R(s,a) + \gamma V(s')]]$$

The robot selects the action with the highest value. Obstacles receive large negative rewards, while reaching the destination receives a positive reward. Value iteration helps the robot learn an efficient shortest path.

2. Recommendation System – Monte Carlo Control

Monte Carlo Control learns from complete customer sessions.

- State: User history and current preferences.
- Action: Recommend a product/movie/song.
- Reward: Positive for clicking or purchasing; negative for ignoring.

After each session, the total return is calculated and action values are updated:

$$[\quad Q(s,a) \leftarrow Q(s,a) + \alpha [G - Q(s,a)]$$

After many sessions, the system learns which recommendations produce higher rewards and improves its policy.

3. Delivery Robot – SARSA

SARSA is an on-policy RL algorithm. The robot observes state S, selects action A, receives reward R, and observes the next state and action.

$$[\quad Q(S,A) \leftarrow Q(S,A) + \alpha [R + \gamma Q(S',A') - Q(S,A)]$$

The robot learns safe delivery paths. Since SARSA learns from the actions it actually follows, its policy can adapt when obstacles or environmental conditions change.

4. Robotic Arm – REINFORCE

REINFORCE is a policy-gradient algorithm suitable for learning a policy directly.

- State: Arm position, object position, and movement information.
- Action: Joint movements.
- Reward: Positive for successful manipulation and negative for errors.

The policy is updated using:

$$[\quad \theta \leftarrow \theta + \alpha G_t \nabla_{\theta} \log \pi_{\theta}(a_t | s_t)$$

Repeated training increases the probability of actions that successfully complete the task.

5. Q-Learning vs DQN for Autonomous Vehicles

Feature	Q-Learning	DQN
State representation	Small/discrete	Large/complex
Learning	Q-table	Neural network
Scalability	Low	High
Convergence	Fast for small problems	Slower but scalable
Driving environments	Limited	More suitable

Conclusion: Q-Learning works well for simple environments, while DQN is better for complex autonomous-driving environments with large state spaces.

6. DQN vs DDQN vs Dueling DQN

Standard DQN: Uses one network to estimate Q-values but can overestimate action values.

DDQN: Separates action selection from action evaluation, reducing overestimation bias and improving learning stability.

Dueling DQN: Separates the network into:

- State-value function ($V(s)$)
- Advantage function ($A(s,a)$)

It learns which states are valuable even when different actions have similar effects.
Conclusion: DDQN improves value estimation, while Dueling DQN improves learning efficiency and representation.

7. A2C vs A3C for Drone Navigation

A2C (Advantage Actor-Critic) uses multiple agents/workers but updates them synchronously.

A3C (Asynchronous Advantage Actor-Critic) uses multiple workers that update the shared model asynchronously.

- Parallel learning collects experience faster.
- A3C can reduce correlation between experiences.
- A2C provides more stable and synchronized updates.
- A3C can achieve faster learning but may have greater implementation complexity.

Conclusion: Both improve learning through parallel experience, with A2C generally offering more stable and predictable updates.

8. POMDP in Healthcare

A standard MDP assumes the complete patient state is known. In healthcare, some information may be missing or uncertain.

A POMDP handles this using observations and a belief state, which represents the probability of different patient conditions.

Example:

Hidden state → Patient's actual condition

Observation → Symptoms, test results, vital signs

The system updates its belief as new information arrives and selects better treatment decisions.

Thus, POMDP is more suitable for healthcare environments with incomplete information.

9. PPO vs TRPO vs DDPG

Algorithm	Stability	Sample Efficiency	Complexity	Suitable For
PPO	High	Good	Medium	Continuous control
TRPO	Very High	Good	High	Stable policy learning
DDPG	Moderate	High	Medium	Continuous actions

PPO: Simple and stable, making it a strong general choice.

TRPO: Provides strong policy-update stability but requires more computation.

DDPG: Handles continuous actions effectively but can be sensitive to hyperparameters.

Conclusion: PPO is generally a good balance between stability, efficiency, and computational complexity.

10. MARL for Smart City Traffic Management

MARL uses multiple RL agents, where each traffic signal acts as an agent.

- State: Traffic density, queue length, waiting time.
- Action: Change or maintain signal.
- Reward: Reduced congestion and waiting time.

Advantages

- Coordinates multiple intersections.
- Improves traffic flow.
- Adapts to changing traffic conditions.

Challenges

- Large number of agents increases complexity.
- Communication and coordination can be difficult.
- Traffic priorities must be fair.
- Emergency vehicles and pedestrians must receive priority.
- Privacy and safety must be considered.

Conclusion: MARL can significantly improve traffic management, but scalability, fairness, coordination, and ethical constraints must be incorporated into the system.

Code of Conduct:

I _G.Srivardhan(192425208)_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: G.Srivardhan

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Conceptual Understanding	25	Demonstrates deep understanding of concepts with complete clarity	Explains concepts correctly with minor gaps	Partial understanding with errors or omissions	Unable to explain basic concepts
Application	25	Effectively applies concepts to new situations; solves problems logically	Applies concepts with minor errors	Limited ability to apply concepts; requires guidance	Unable to apply concepts effectively
Analytical Thinking	20	Critically analyzes scenarios with strong justification and reasoning	Provides reasonable analysis with some justification	Superficial analysis with weak reasoning	No analytical reasoning demonstrated

Communication	15	Communicates ideas clearly, confidently, and with appropriate terminology	Communicates clearly but with minor hesitation	Communication lacks clarity or structure	Unable to communicate ideas effectively
Response to Questions	15	Responds accurately and confidently, extending answers with depth	Responds correctly but with limited depth	Responds partially; requires prompting	Unable to respond to questions effectively